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Enhanced credit risk prediction using deep learning and SMOTE-ENN resampling

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ABSTRACT

Credit risk prediction is a vital task in financial services, ensuring that institutions can manage their lending risks effectively. This study investigates the effectiveness of deep learning (DL) models for credit risk prediction, with a focus on addressing the challenge of class imbalance and the black box nature of these models using the Synthetic Minority Over-sampling Technique - Edited Nearest Neighbor (SMOTE-ENN) resampling method and Shapley Additive Explanations (SHAP), respectively. The study compares the performance of various DL architectures, including Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), Gated Recurrent Units (GRU), and Graph Neural Networks (GNN), on two real-world datasets: the Australian and German credit datasets. The findings reveal that the GRU model, enhanced with SMOTE-ENN resampling, outperforms other models in terms of accuracy, sensitivity, and specificity. The superior performance of the GRU-SMOTE-ENN model demonstrates its potential as a robust deep learning technique for financial institutions to enhance credit risk assessment. Additionally, the study demonstrates how the integration of SHAP values significantly improves the interpretability of deep learning models, making them more transparent and trustworthy for stakeholders.

1. Introduction

Credit risk assessment is crucial in the financial sector, directly influencing decisions on loans, credit cards, and other forms of credit issuance (Bao et al., 2019; Li & Chen, 2020; Machado & Karray, 2022). Annually, companies lose billions of dollars due to poor credit decisions, demonstrating its significant economic impact (Alagic et al., 2024; Aruleba & Sun, 2025; Wong et al., 2023). Traditional models for assessing credit risk, such as logistic regression and decision trees, though easy to understand and interpret, often fail to capture complex nonlinear patterns and may not effectively handle large and diverse datasets (Addo et al., 2018; Bao et al., 2019). These limitations can lead to suboptimal decision-making.

Deep learning (DL) offers a substantial improvement over traditional machine learning (ML) models in processing complex and voluminous data. These models employ sophisticated architectures like convolutional neural networks (CNN) and recurrent neural networks (RNNs) that excel in identifying and learning patterns at multiple levels of abstraction (Kattenborn et al., 2021; Khagi & Kwon, 2022). This capability makes them highly effective for dynamic and detailed credit risk assessments.

A major challenge in credit risk prediction is the imbalanced nature of datasets, where instances of default are significantly outnumbered by non-default cases. This imbalance can bias predictive models towards the majority class, leading to inadequate risk assessment (Rao et al., 2023). To address this, the use of Synthetic Minority Over-sampling Technique-Edited Nearest Neighbors (SMOTE-ENN) for resampling balances the dataset by synthesizing new minority class samples and cleaning overlapping values, thereby enhancing model training and performance (Bounab et al., 2024).

This study is focused on a detailed analysis of how deep learning models enhanced with SMOTE-ENN can improve credit risk prediction. It evaluates the performance of various deep learning architectures such as CNN, Long Short-Term Memory networks (LSTM), Gated Recurrent Units (GRU), and Graph Neural Networks (GNN). To ensure the transparency and interpretability of the predictive models, the SHapley Additive exPlanations (SHAP) technique will be utilized. SHAP values provide insights into the contribution of each feature to the prediction, making it easier to interpret the complex decision-making processes of deep learning models (Antwarg et al., 2021; Mienye et al., 2024; Yang et al., 2021).

The main goal and contributions of this study are outlined as follows:

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- A comprehensive comparison of multiple deep learning models to establish a performance benchmark in credit risk prediction.
- An evaluation of SMOTE-ENN resampling's effectiveness in managing imbalanced datasets, and enhancing the performance of DL models.
- The use of SHAP for model explanation, aiding in the interpretability of complex models, which is vital for regulatory compliance and stakeholder trust.

The rest of the paper is structured as follows: Section 2 presents a background and related works. Section 3 discusses the datasets, DL architectures, and performance evaluation metrics used in the study. Section 4 presents the proposed credit risk prediction framework, and Section 5 presents and discusses the experimental results. Section 6 concludes the study and highlights future research directions.

2. Related works

Credit risk models have been integral to the financial industry for decades, traditionally relying on statistical methods and ML algorithms like logistic regression, support vector machines (SVM), and decision trees (Alam et al., 2020; Shilpa et al., 2023; Suhadolnik et al., 2023). These algorithms, especially decision trees, are known for their simplicity and interpretability (Mienye & Jere, 2024c). Despite their widespread use, traditional methods struggle with complex, non-linear relationships in data, often leading to less accurate credit risk predictions when dealing with large and high-dimensional datasets (Birla et al., 2016; Wang, 2022).

Meanwhile, the evolution of big data and advanced computational technologies has shifted the focus towards advanced machine learning as an alternative to classical credit scoring techniques. Modern ML techniques such as random forests, gradient boosting machines, and extreme gradient boosting (XGBoost) have shown improved performance in credit risk prediction (Aysan et al., 2024; Emmanuel et al., 2024; Liu et al., 2024). However, these methods also face challenges related to overfitting and limited interpretability, often requiring extensive data preparation and parameter tuning (Mienye & Jere, 2024b).

Deep learning, a further development within machine learning, has received notable attention for being effective in modeling complex relationships within extensive datasets. In the financial sector, deep learning is applied to tasks such as fraud detection, algorithmic trading, and risk management. For credit risk prediction, models like CNNs and recurrent neural networks (RNNs) have demonstrated enhanced performance over other ML models by efficiently processing sequential and spatial data (Mahbobi et al., 2021).

Despite these advancements, deep learning models often suffer from a lack of interpretability, which is a critical issue for applications within regulated industries like finance (Nazemi & Fabozzi, 2024). The opaque nature of these models can hinder regulatory compliance and reduce trust among stakeholders (Lisboa et al., 2023; Talaat et al., 2024). Additionally, the problem of imbalanced datasets in credit risk prediction—where default cases are far fewer than non-default cases—remains a significant challenge. Meanwhile, oversampling and undersampling techniques have been mainly used to balance the imbalance of credit risk prediction models (Abedin et al., 2023; Lenka et al., 2023; Pan & Yang, 2023). Traditional resampling techniques such as oversampling can lead to model overfitting by artificially increasing the minority class, while undersampling can eliminate crucial information by reducing the majority class (Vairetti et al., 2024).

To overcome the imbalance class problem, this study uses the SMOTE-ENN, which combines SMOTE with ENN cleaning to not only augment the minority class through synthetic examples but also remove overlapping instances that could degrade model performance (Bounab et al., 2024). This approach helps in balancing classes effectively without losing significant data points.

Additionally, the study applies deep learning architectures for credit risk prediction and the SHAP technique for model interpretation. The study will critically evaluate the performance of architectures such as CNNs, LSTM, GRU, GNN, and MLP. By exploring the comparative effectiveness of these models, the research will provide insights into their applicability and effectiveness in credit risk prediction, setting the stage for future advances in financial analytics.

3. Materials and methods

3.1. Datasets

In this study, we utilize two widely recognized datasets for evaluating the performance of ML models in credit risk prediction: the Australian credit dataset and the German credit dataset. The Australian credit dataset, shown in Table 1, comprises various personal and financial attributes of loan applicants (Pławiak et al., 2019). This dataset is frequently utilized in credit risk studies due to its comprehensive set of features. The dataset is characterized by its 15 distinct features, which cover a wide range of attributes such as demographic information, credit history, and financial details, including the outcome of the credit decision. The dataset exhibits a notable class imbalance, with 44.49% of the samples belonging to the minority class (bad credit risk).

The German credit dataset (Table 2) includes information about the credit history, loan purpose, credit amount, and personal details of applicants (Boughaci et al., 2021). This dataset is well-known for its detailed feature set and has been extensively used in academic research to evaluate the effectiveness of different credit scoring algorithms. The features are both categorical and numerical, providing a diverse set of variables for model training and evaluation. Similar to the Australian dataset, the German credit dataset also displays class imbalance, with 30% of the samples representing the minority class (bad credit risk).

3.2. Deep learning architectures

3.2.1. Multilayer Perceptron

A Multilayer Perceptron (MLP) is a class of feedforward artificial neural network. It consists of at least three layers of nodes: an input layer, a hidden layer, and an output layer. Except for the input nodes, each node is a neuron that uses a nonlinear activation function. The purpose of these layers is to progressively transform the input data into a meaningful output through a series of linear and nonlinear operations. Mathematically, an MLP can be represented as follows:

$$\mathbf{h}^{(1)} = \sigma(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \tag{1}$$

$$\mathbf{h}^{(2)} = \sigma(\mathbf{W}^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \tag{2}$$

$$\mathbf{y} = \sigma(\mathbf{W}^{(3)}\mathbf{h}^{(2)} + \mathbf{b}^{(3)}) \tag{3}$$

where $\mathbf{x}$ is the input vector, $\mathbf{W}$ and $\mathbf{b}$ are the weights and biases for each layer, σ is the activation function, $\mathbf{h}$ denotes the hidden layers, and $\mathbf{y}$ is the output vector. The weights and biases are learned during the training process to minimize the difference between the predicted output and the actual target values.

3.2.2. Convolutional Neural Networks

CNNs are designed to process data with grid-like topology, such as images. A CNN consists of convolutional layers, pooling layers, and fully connected layers (Krizhevsky et al., 2017; Yamashita et al., 2018). Convolutional layers apply a series of filters to the input data to extract spatial features, while pooling layers reduce the spatial dimensions, thereby decreasing the computational load and controlling overfitting.

A convolution operation on a matrix $\mathbf{X}$ with a filter $\mathbf{F}$ is defined as:

$$(\mathbf{X} * \mathbf{F})(i, j) = \sum_m \sum_n \mathbf{X}(i + m, j + n)\mathbf{F}(m, n) \tag{4}$$

where $\mathbf{X}$ is the input, $\mathbf{F}$ is the filter, and (i, j) are the indices of the output matrix. This operation produces a feature map that highlights important features in the input data, which are then passed through activation functions and pooling layers for further processing.

Table 1
Australian credit dataset.

S/N	Feature	Description	Data Type
1	A1	Status of existing checking account	Categorical
2	A2	Duration of credit (months)	Numeric
3	A3	Credit history	Categorical
4	A4	Purpose of the credit	Categorical
5	A5	Credit amount	Numeric
6	A6	Savings account/bonds	Categorical
7	A7	Present employment duration	Categorical
8	A8	Installment rate as a percentage of disposable income	Numeric
9	A9	Personal status and sex	Categorical
10	A10	Other debtors/guarantors	Categorical
11	A11	Present residence duration	Numeric
12	A12	Property	Categorical
13	A13	Age in years	Numeric
14	A14	Other installment plans	Categorical
15	Class	Credit risk (1 = good, 2 = bad)	Binary

Table 2
German credit dataset.

S/N	Feature	Description	Data Type
1	Status of existing checking account	Status of the applicant's checking account	Categorical
2	Duration	Length of the credit in months	Numeric
3	Purpose	Reason for the credit request	Categorical
4	Credit amount	Amount of credit requested by the applicant	Numeric
5	Savings account	Status of the applicant's savings account	Categorical
6	Age	Age of the applicant in years	Numeric
7	Employment	Duration of the applicant's employment	Categorical
8	Sex	Gender of the applicant	Categorical
9	Housing	Applicant's housing status (e.g., own/rent/free)	Categorical
10	Job	Category of the applicant's job	Categorical
11	Credit Risk	Class variable indicating credit risk (1 = good, 2 = bad)	Binary

3.2.3. Long Short-Term Memory Networks

LSTMs are a type of recurrent neural network designed to capture long-term dependencies by incorporating memory cells and gating mechanisms (Zheng et al., 2017). These mechanisms regulate the flow of information, making LSTMs more effective at capturing long-range dependencies than standard RNNs. The LSTM cell is defined by the following equations:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \tag{5}$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \tag{6}$$

$$\mathbf{C}_t = \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_C[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_C) \tag{7}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \tag{8}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{C}_t) \tag{9}$$

where $\mathbf{f}_t$ is the forget gate, $\mathbf{i}_t$ is the input gate, $\mathbf{C}_t$ is the cell state, $\mathbf{o}_t$ is the output gate, $\mathbf{h}_t$ is the hidden state, $\mathbf{x}_t$ is the input, $\mathbf{W}$ and $\mathbf{b}$ are weights and biases, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication (Esenogho et al., 2022). The forget gate $\mathbf{f}_t$ controls what information to discard from the cell state, while the input gate $\mathbf{i}_t$ controls what new information to add. The output gate $\mathbf{o}_t$ determines the output based on the cell state. Fig. 1 shows the LSTM architecture.

3.2.4. Gated Recurrent Units

GRUs are a simplified version of LSTMs that combine the forget and input gates into a single update gate (Cahuantzi et al., 2023; Mienye & Jere, 2024a). This simplification makes GRUs computationally efficient while still retaining the ability to capture long-term dependencies. The GRU cell is defined by:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_z) \tag{10}$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_r) \tag{11}$$

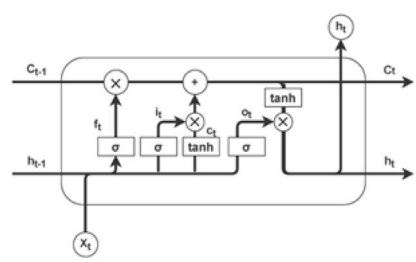

Fig. 1. LSTM architecture (Mienye & Jere, 2024a).

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}[\mathbf{r}_t \odot \mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}) \tag{12}$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \tag{13}$$

where $\mathbf{z}_t$ is the update gate, $\mathbf{r}_t$ is the reset gate, $\tilde{\mathbf{h}}_t$ is the candidate activation, $\mathbf{h}_t$ is the hidden state, $\mathbf{x}_t$ is the input, $\mathbf{W}$ and $\mathbf{b}$ are weights and biases, and σ is the sigmoid function. The update gate $\mathbf{z}_t$ controls the degree to which the previous state $\mathbf{h}_{t-1}$ is maintained, while the reset gate $\mathbf{r}_t$ determines how much of the past state to forget (Mienye & Jere, 2024a). The GRU architecture is shown in Fig. 2.

3.2.5. Graph Neural Networks

GNNs are designed to process data that is structured as graphs. They generalize neural networks to work with graph data by aggregating information from neighboring nodes (Khemani et al., 2024; Zhou et al., 2020). This makes GNNs suitable for tasks involving relational data, such as social network analysis or molecular structure prediction. However, they have been applied in different binary classification problems, achieving excellent performance (Feng et al., 2022). A GNN layer can be defined as:

$$\mathbf{h}_v^{(k)} = \sigma\left(\mathbf{W}^{(k)} \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(k-1)} + \mathbf{b}^{(k)}\right) \tag{14}$$

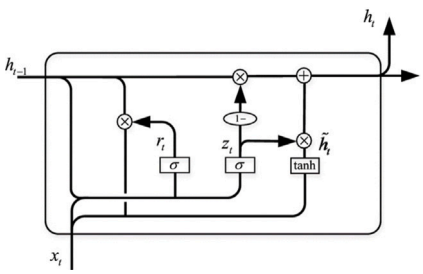

Fig. 2. GRU architecture (Mienye & Jere, 2024a).

where $\mathbf{h}_v^{(k)}$ is the representation of node v at layer k, $\mathcal{N}(v)$ is the set of neighbors of node v, $\mathbf{W}^{(k)}$ and $\mathbf{b}^{(k)}$ are the weight matrix and bias at layer k, and σ is the activation function (Khemani et al., 2024). The aggregation function combines information from a node's neighbors to update its representation, capturing the graph's structural information.

3.3. SMOTE-ENN

The SMOTE-ENN is a hybrid resampling method that effectively addresses class imbalance by combining oversampling and noise reduction techniques. SMOTE generates synthetic samples for the minority class by interpolating between existing minority instances and their k-nearest neighbors (Mienye & Sun, 2023). Mathematically, for a given minority class instance x, a synthetic sample $x_{\text{synthetic}}$ is generated as:

$$x_{\text{synthetic}} = x + \delta \cdot (x_{knn} - x), \tag{15}$$

where $\delta \in [0, 1]$ is a random value and x_{knn} is one of the k-nearest neighbors of x. This process helps balance the class distribution, reducing bias in the model's predictions. However, SMOTE alone may introduce noise, as it does not account for overlapping or outlier instances in the dataset (Nnamoko & Korkontzelos, 2020). To mitigate this issue, the Edited Nearest Neighbor (ENN) algorithm is applied after SMOTE. ENN removes instances whose class labels differ from the majority class of their k-nearest neighbors (Kanj et al., 2016). Formally, for each instance x in the dataset, let $\mathcal{N}_k(x)$ denote the set of k-nearest neighbors of x. ENN removes x if:

$$\text{MajorityClass}(\mathcal{N}_k(x)) \neq \text{Class}(x). \tag{16}$$

This cleaning step reduces noise and improves the quality of the training data by removing mislabeled or ambiguous samples. SMOTE-ENN integrates these two techniques sequentially. First, SMOTE generates synthetic samples to balance the dataset. Then, ENN refines the combined dataset (original + synthetic) by removing noisy and overlapping samples (Mienye & Sun, 2023). This dual approach ensures that the resampled dataset is both balanced and clean, enhancing the model's ability to generalize.

SMOTE-ENN was chosen for this study due to its ability to address key limitations of other resampling techniques. While SMOTE alone may oversample noisy or overlapping regions, and alternatives such as SMOTE-SVM may overfit to decision boundaries, SMOTE-ENN strikes a balance by generating synthetic samples and subsequently cleaning the dataset. This combination is particularly advantageous for datasets with high class imbalance and noisy samples, as it ensures a cleaner and more representative training set. Furthermore, for deep learning models such as GRUs, which are sensitive to noise and imbalance, SMOTE-ENN provides a robust preprocessing framework. By improving the representation of the minority class while removing noise, SMOTE-ENN enhances the model's ability to learn meaningful patterns, ensuring reliable and fair predictions. This preprocessing step is particularly critical in credit risk prediction, where imbalanced datasets are common, and model performance on the minority class is crucial for decision-making.

The research carried out by Han and Joe (2024) showed that SMOTE-ENN helps to reduce bias and variance in model. When a model is trained with a dataset that is balanced, there is a high probability that the model will be predicting majorly for the classes with the highest majority which might make the model to have a higher accuracy but the recall score for the minority classes will be poor (Johnson & Khoshgoftaar, 2019). Hence, to balance the dataset, and to avoid any issue in regards to the performance of the model, SMOTE-ENN is utilized (Puri & Kumar Gupta, 2022).

Also, to provide insights to the decision-making process of the models, explainability techniques such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) help to explain the model (Parisineni & Pal, 2024). Moreover, SMOTE-ENN helps to reduce the computational issues that might come up while training with a large and noisy dataset (Nishat et al., 2022). It also removes samples that are not needed,especially while training deep learning models (Munappy et al., 2022). Furthermore, SMOTE-ENN is applied to both static tabular data and dynamic datasets, which makes it useful in other applications like anomaly detection and fraud detection (Tejeiro Koller, 2016).

3.4. Evaluation metrics

To evaluate the performance of the DL models in this study, the following metrics are used to provide a comprehensive assessment: accuracy, sensitivity, and specificity. Accuracy measures the proportion of true results (both true positives and true negatives) among the total number of cases examined (Obaido et al., 2024). It is a commonly used metric for classification tasks and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives. However, accuracy alone can be misleading, especially in cases of imbalanced datasets, where the number of true negatives significantly outweighs the number of true positives (Mienye & Jere, 2024c). To address this, sensitivity and specificity are also considered. Sensitivity, also known as recall or the true positive rate, measures the proportion of true positives among the actual positives (TP). It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{18}$$

Meanwhile, specificity, or true negative rate, complements sensitivity by measuring the proportion of true negatives among the actual negatives (Trevethan, 2017). It is defined as:

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{19}$$

4. Proposed credit risk prediction framework

The proposed framework for credit risk prediction utilizes advanced DL models coupled with the SMOTE-ENN resampling technique to address the challenge of class imbalance prevalent in credit datasets. The framework, described in Algorithm 1, involves data preprocessing, model training, evaluation, and explanation, thereby enhancing both predictive accuracy and model transparency.

This proposed framework employs SMOTE-ENN to balance the data and mitigate bias towards the majority class, thereby improving the fairness and accuracy of predictions across diverse applicant profiles. Secondly, the use of DL models is robust for capturing complex, non-linear interactions within the data, which are often missed by traditional ML models. Thirdly, by incorporating SHAP for interpretation, the framework addresses the critical need for transparency and explainability in financial decision-making systems.

Algorithm 1 Deep Learning-Based Credit Risk Prediction Framework

Input: Training data $\mathcal{D}_{train}$ and testing data $\mathcal{D}_{test}$
Output: Credit risk predictions $\mathcal{P}$
Step 1: Data Preprocessing
Normalize and encode features in $\mathcal{D}_{train}$ and $\mathcal{D}_{test}$.
Apply SMOTE-ENN to $\mathcal{D}_{train}$ to balance the classes, enhancing the representation of minority classes.
Step 2: Train Deep Learning Models
Train the following models on the balanced $\mathcal{D}_{train}$:

1. Convolutional Neural Network (CNN)
2. Long Short-Term Memory network (LSTM)
3. Gated Recurrent Unit (GRU)
4. Graph Neural Network (GNN)

Step 3: Model Evaluation
Evaluate each model using k-fold cross-validation on $\mathcal{D}_{test}$.
Assess models based on accuracy, sensitivity, and specificity to determine the best performer.
Step 4: Interpretation with SHAP Values
Identify the best-performing model.
Apply SHAP values to this model to quantify the impact of each feature on the prediction outcomes, improving interpretability and transparency in decision-making.

Table 3
Hyperparameter settings for different deep learning models.

Hyperparameter	MLP	CNN	LSTM	GRU	GNN
Learning Rate	0.001	0.0005	0.001	0.001	0.0005
Batch Size	32	64	64	64	32
Hidden Units	128	256	256	256	128
Dropout	0.3	0.4	0.3	0.3	0.2
Optimizer	Adam	Adam	RMSprop	Adam	AdamW
Epochs	100	100	100	100	80

5. Results and discussion

The experimental results for the different models are presented and discussed in this section. To ensure reliable and unbiased performance evaluation, a 10-fold cross-validation technique was employed. In this approach, the dataset is randomly split into 10 equal parts (or folds). Each model is trained on 9 folds and tested on the remaining 1 fold, and this process is repeated 10 times, each time using a different fold as the test set. The final performance metrics are then obtained by averaging the results across all 10 iterations. This method helps mitigate the potential variability caused by a single train–test split and provides a more robust assessment of model performance.

Additionally, different random seeds were used to generate multiple training and test sets, ensuring that the models' performance metrics reflect an average across various splits. The models were trained using the original datasets and the resampled data, allowing for a comparison of performance before and after introducing the SMOTE-ENN resampling. Meanwhile, for the different models, the hyperparameters used are tabulated in Table 3.

To ensure optimal performance across the deep learning models, the hyperparameter values were determined through a combination of hyperparameter tuning using grid search, reference to recent studies, and default values for well-established architectures. A grid search approach was employed to fine-tune critical parameters, such as learning rate, batch size, and dropout rate, by evaluating multiple configurations and selecting those yielding the highest validation accuracy and lowest generalization error.

For recurrent models (LSTM, GRU), hyperparameter choices were guided by recent studies on credit risk prediction and financial time-series modeling, where a learning rate of 0.001, batch size of 64, and dropout rate of 0.3 have been shown to stabilize training and prevent overfitting. CNN and GNN hyperparameters were similarly optimized, with lower learning rates (0.0005) chosen based on prior research to accommodate their sensitivity to rapid weight updates.

The number of hidden units was adjusted based on model complexity, with MLP and GNN using 128 units and LSTM, GRU, and CNN using 256 units to enhance representational capacity. The Adam optimizer was selected for most models due to its adaptive moment estimation, while RMSprop was preferred for LSTM, following best practices in handling long-range dependencies. By combining empirical tuning and insights from recent literature, this selection ensures robust and well-generalized model performance across different credit risk datasets.

Table 4
Classifiers performance on Australian dataset before resampling.

Classifier	Accuracy	Sensitivity	Specificity
MLP	0.810	0.850	0.781
CNN	0.859	0.860	0.885
LSTM	0.861	0.848	0.885
GRU	0.837	0.829	0.849
GNN	0.890	0.868	0.886

Table 5
Classifiers performance on Australian dataset after resampling.

Classifier	Accuracy	Sensitivity	Specificity
MLP	0.850	0.879	0.830
CNN	0.876	0.901	0.908
LSTM	0.917	0.897	0.913
GRU	0.926	0.911	0.930
GNN	0.905	0.895	0.918

5.1. Model performance using the Australian dataset

The results of classifier performance on the Australian dataset, as shown in Tables 4 and 5, demonstrate notable improvements in accuracy, sensitivity, and specificity after resampling. These enhancements indicate the effectiveness of resampling techniques in addressing class imbalance, a prevalent issue in credit risk datasets. The resampling ensured a more balanced dataset, which, in turn, improved the learning process.

Among the models tested, the GRU stands out as the best-performing model post-resampling, achieving the highest accuracy at 0.926, with sensitivity and specificity scores of 0.911 and 0.930, respectively. This performance highlights the GRU's robustness in learning from the balanced credit datasets. Meanwhile, it can also be observed that the performance of the various DL models significantly improved after the SMOTE-ENN resampling. These outcomes highlight the critical role of deep learning models in improving predictive accuracy in credit risk assessment. Additionally, the robust performance of models such as the GRU, following resampling, suggests their strong potential to provide more reliable and fair credit risk prediction outcomes.

Furthermore, the SHAP summary plot (Fig. 3) for the GRU model on the Australian dataset reveals distinct patterns in how different features influence credit risk prediction. Notably, features A9 and A15 show a strong positive influence when their values are high, indicating a lower risk of default. Conversely, features A2 and A5 exhibit a predominantly negative impact for larger values. Features A3 and A6 display a mix of positive and negative impacts on the model output, reflecting the subtle consideration of these factors in assessing creditworthiness.

5.2. Model performance using the German dataset

Performance assessments of various classifiers on the German dataset, as detailed in Tables 6 and 7, reveal significant enhancements in accuracy, sensitivity, and specificity following resampling.

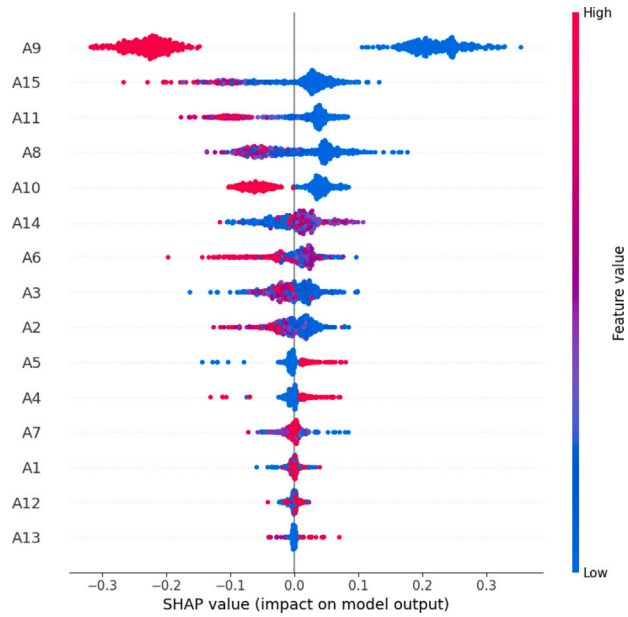

Fig. 3. GRU model explanation on the Australian dataset.

Table 6
Classifiers performance on German dataset before resampling.

Classifier	Accuracy	Sensitivity	Specificity
MLP	0.704	0.773	0.310
CNN	0.751	0.782	0.399
LSTM	0.730	0.770	0.394
GRU	0.749	0.895	0.431
GNN	0.767	0.790	0.446

Table 7
Classifiers performance on German dataset after resampling.

Classifier	Accuracy	Sensitivity	Specificity
MLP	0.824	0.830	0.810
CNN	0.791	0.828	0.709
LSTM	0.819	0.859	0.729
GRU	0.899	0.931	0.852
GNN	0.860	0.910	0.830

This improvement highlights the importance of resampling methods in addressing the class imbalance, which is prevalent in credit scoring datasets. Effective resampling enables the models to better generalize from the data, leading to more accurate predictions.

Using the German dataset, the GRU still displays superior performance after resampling, achieving an accuracy of 0.899, with sensitivity and specificity rates of 0.931 and 0.852, respectively. These results demonstrate the GRU's effectiveness in processing the credit risk dataset. These findings show the vital role of DL models in enhancing predictive performance in financial applications. The notable improvement observed with the GRU model post-resampling suggests its potential to deliver more reliable and equitable outcomes in credit risk prediction. Meanwhile, Fig. 4 shows the GRU model interpretation.

Notably, the feature "Checking account" shows a high dispersion of SHAP values, indicating a significant influence on model output, where higher values in the checking account lead to a more positive impact on credit risk assessment. In contrast, "Credit amount" and "Duration" exhibit primarily negative impacts, suggesting that higher values for these features increase the likelihood of a negative credit decision. Furthermore, age and job appear to have a moderate influence with mixed effects on the model's output. The spread of SHAP values across features like "Purpose" and "Housing" suggests variability in how they

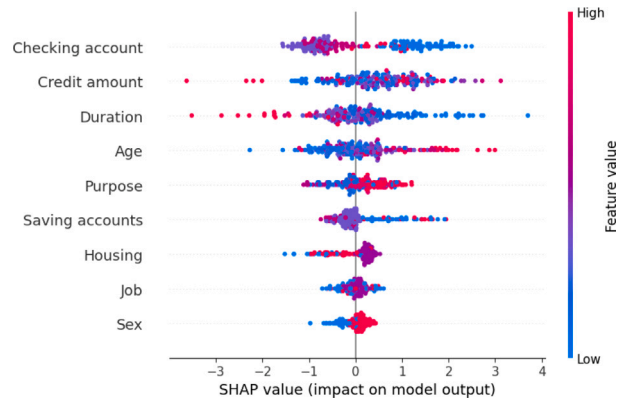

Fig. 4. GRU model explanation on the German dataset.

```
Please provide the following details:
Age: 34
Sex (male/female): male
Job (0-3): 0
Housing (own/free/rent): own
Saving accounts (little/moderate/rich/quite rich): little
Checking account (little/moderate/rich): rich
Credit amount: 4000
Duration (in months): 4
Purpose (car/furniture/equipment/radio/TV/education/business/domestic appliances/repairs/vacation/others): car
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 561ms/step

Prediction Score: 0.0000
Predicted Class: Bad Risk
```

Fig. 5. The real time prediction of the credit score (Bad risk).

```
Please provide the following details:
Age: 49
Sex (male/female): male
Job (0-3): 1
Housing (own/free/rent): own
Saving accounts (little/moderate/rich/quite rich): little
Checking account (little/moderate/rich): little
Credit amount: 2096
Duration (in months): 12
Purpose (car/furniture/equipment/radio/TV/education/business/domestic appliances/repairs/vacation/others): education
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 550ms/step

Prediction Score: 1.0000
Predicted Class: Good Risk
```

Fig. 6. The real time prediction of the credit score (Good risk).

are weighted by the model, underlining the complexity of interactions within the data that GRU models capably capture.

Different SMOTE versions were applied on the German dataset. The GRU model achieved an accuracy of 72%. Meanwhile, other models such as the CNN, which was trained before resampling with SMOTE, performed better with an accuracy of 76.5%. However, the main focus in this section was to evaluate the models after resampling in order to ensure that the dataset was balanced. The findings showed that models trained before resampling outperformed those trained after resampling, with the CNN model achieving the highest accuracy. Among the resampling techniques applied, SMOTE alone yielded better performance compared to more complex versions like BorderlineSMOTE, SMOTEENN, and SMOTETomek. Notably, the SMOTEENN technique significantly reduced the model accuracy, especially for the LSTM models, which had performance as low as 50%. This indicates that for this particular dataset, simple oversampling using SMOTE was more effective than the aggressive resampling techniques, which introduced noise and negatively affected model performance. In conclusion, the GRU models adapted well to resampling, while CNN models remained the strongest performers without resampling.

The performance of the GRU-SMOTE-ENN model had a positive impact. It directly contributes to the improvement in sensitivity and specificity and also helped to avoid class imbalance in the dataset. It also helps to ensure that the minority classes are seen by the model. The GRU-SMOTE-ENN was implemented in real-time on a command line application to assess and predict the credit risk of a user. Fig. 5 shows the real time prediction of the credit score (Bad risk) while Fig. 6 shows the real time prediction of the credit score (Good risk).

Table 8
Classifiers performance on European dataset before resampling.

Classifier	Accuracy	Sensitivity	Specificity
MLP	0.986	0.741	0.896
CNN	0.989	0.788	0.899
LSTM	0.991	0.836	0.924
GRU	0.990	0.822	0.910
GNN	0.989	0.815	0.803

Table 9
Classifiers performance on European dataset after resampling.

Classifier	Accuracy	Sensitivity	Specificity
MLP	0.993	0.944	0.960
CNN	0.998	0.957	0.972
LSTM	0.999	0.976	0.992
GRU	0.999	0.953	0.971
GNN	0.994	0.936	0.958

5.3. Model performance using the European credit dataset

The German and Australian datasets used to evaluate the performance of the models have 1,000 and 690 instances, respectively. While these datasets are useful for small-scale model evaluation, they may not fully capture the complexities and class imbalances found in larger, real-world financial datasets. Therefore, the European credit card dataset, containing 30 features and approximately 284,807 transactions collected over two days in September 2013, is employed to further validate the robustness of the proposed approach. Of these transactions, only 492 are fraudulent, representing a highly imbalanced dataset with only 0.17% belonging to the minority (fraudulent) class.

Table 8 shows the performance of the classifiers before resampling, where we observe significant issues in sensitivity across all models. This indicates that the models had difficulty predicting the minority class effectively. In particular, the MLP model only achieved a sensitivity of 0.741, while the LSTM performed slightly better with a sensitivity of 0.836. However, accuracy and specificity remained relatively high, suggesting that the models were biased towards predicting the majority class.

After applying the SMOTE-ENN resampling technique, the models showed significant improvement in their ability to predict the minority class. As seen in Table 9, the LSTM model now achieved a sensitivity of 0.976, while maintaining a high accuracy of 0.999 and a specificity of 0.992. This indicates that resampling not only addressed the class imbalance but also improved the overall performance of the models across all metrics.

Furthermore, the interpretability of the LSTM model is enhanced through SHAP values, as shown in Fig. 7. The SHAP summary plot provides a comprehensive explanation of how each feature impacts the LSTM model's output. For example, feature `V14` has the highest SHAP value, indicating that it strongly contributes to the prediction, while features like `Amount` and `Time` also play significant roles. This level of interpretability is crucial in the financial industry, where understanding model decisions is essential for regulatory compliance and stakeholder trust.

5.4. Comparison with other scholarly works

The classifiers' performance on the Australian and German datasets reveals that the DL models, particularly those enhanced with resampling techniques like SMOTE-ENN, significantly outperform traditional methods. Tables 10 and 11 provide a comprehensive comparison. From the Tables, our approach using GRU combined with SMOTE-ENN resampling achieves comparable performance with state-of-the-art techniques.

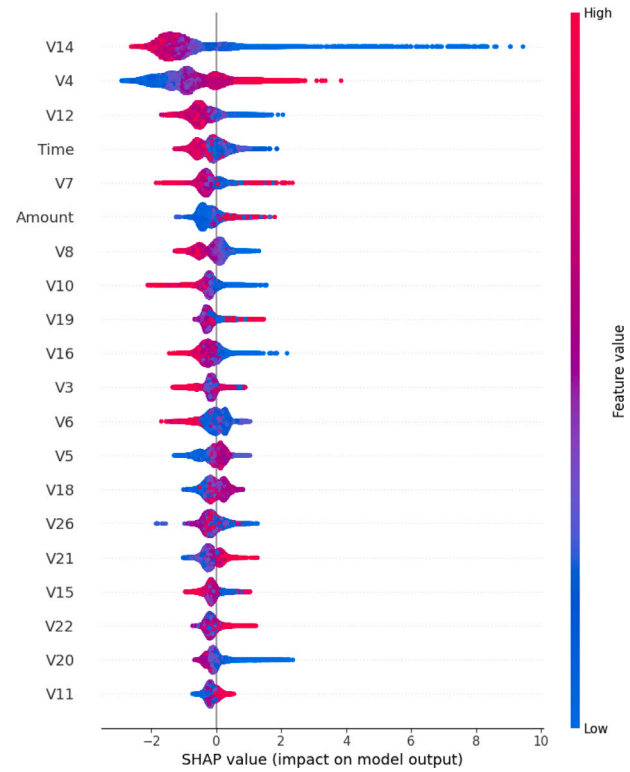

Fig. 7. LSTM model explanation on the European dataset.

In the Australian dataset, our GRU-SMOTE-ENN model, with an accuracy of 0.926, sensitivity of 0.911, and specificity of 0.930, outperformed other methods. Similarly, for the German dataset, the GRU-SMOTE-ENN method stands out with an accuracy of 0.899. These results demonstrate the importance of integrating advanced resampling methods with powerful classifiers like GRUs to tackle prevalent issues such as class imbalance in credit datasets. The superiority of the GRU-SMOTE-ENN combination in our study suggests that it could serve as a valuable model for future credit risk assessment tasks.

The novelty of this work lies in the combination of GRU models with SMOTE-ENN to improve performance on imbalanced datasets.

5.5. Discussion

The results highlighted in the previous sections demonstrate a significant enhancement in model performance due to the application of SMOTE-ENN resampling combined with the GRU model. The GRU-SMOTE-ENN model outperformed traditional ML models and other DL configurations, particularly in metrics critical for reliable credit risk assessment—sensitivity and specificity. This finding is significant as it suggests that deep learning, when integrated with advanced resampling techniques like SMOTE-ENN, can effectively address class imbalance, a common challenge in financial datasets.

The choice of SMOTE-ENN as the resampling technique was motivated by its hybrid approach, which combines the benefits of SMOTE oversampling and ENN undersampling. This method not only generates synthetic minority samples to balance the dataset but also removes noise and overlapping data points, leading to cleaner and more representative training data. The experimental results validate that SMOTE-ENN provides a more robust solution compared to traditional resampling methods, as it enhances the learning process for the deep learning models without introducing significant overfitting.

Moreover, the superior performance of the GRU model, especially in credit risk prediction, indicates its capability to identify complex

Table 10
Performance Comparison with other scholarly works that used the Australian dataset.

Reference	Method	Accuracy	Sensitivity	Specificity	Interpretability
Tripathi et al. (2021a)	Random forest with Feature selection	0.873	0.882	0.883	×
Tripathi et al. (2020)	Evolutionary Extreme Learning Machine	0.871	0.859	0.878	×
Lawi et al. (2017)	Boosted Logistic Regression Ensemble	0.880	–	–	×
Lenka et al. (2022)	Decision Tree and SMOTE	0.838	–	–	×
Lenka et al. (2022)	Decision Tree and ADASYN	0.816	–	–	×
Koc et al. (2023)	XGBoost and Grid Search	0.890	–	–	×
Han et al. (2019)	Gaussian Mixture Model	0.867	0.898	–	×
Wang et al. (2012)	Rotation Forest	0.865	–	–	×
Tripathi et al. (2021a)	MLP with Feature selection	0.853	0.854	0.858	×
Wang et al. (2012)	Random Subspace Decision Tree	–	0.869	–	×
Dastile and Celik (2021)	Convolutional Neural Network	0.880	–	–	✓
Tripathi et al. (2021b)	MLP-based Bagging Ensemble	0.865	–	–	×
Tripathi et al. (2018)	CART based Ensemble with Majority Voting	0.913	0.953	0.871	×
Our Approach	GRU-SMOTE-ENN	0.926	0.911	0.930	✓

Table 11
Performance Comparison with other scholarly works that used the German dataset.

Reference	Method	Accuracy	Sensitivity	Specificity	Interpretability
Han et al. (2019)	Gaussian Mixture Model	0.742	0.756	–	×
Tripathi et al. (2018)	CART based Ensemble with Majority Voting	0.772	0.960	0.370	×
Esenogho et al. (2022)	LSTM Ensemble with SMOTE-ENN	0.904	–	–	×
Koc et al. (2023)	MLP and Grid Search	0.810	–	–	×
Koc et al. (2023)	XGBoost and Grid Search	0.816	–	–	×
Hassani et al. (2020)	Optimized Random forest	0.856	0.828	–	×
Alam et al. (2020)	Gradient Boosted Decision Tree and SMOTE	0.824	0.838	–	×
Wang et al. (2012)	Rotation Forest	0.770	–	–	×
Wang et al. (2012)	Random Subspace Decision Tree	0.761	–	–	×
Alam et al. (2020)	Gradient Boosted Decision Tree and SMOTETomek	0.835	0.868	–	×
Lawi et al. (2017)	Boosted Logistic Regression Ensemble	0.810	–	–	×
Tripathi et al. (2021a)	MLP with Feature selection	0.761	0.691	0.700	×
Gicić et al. (2023)	three-layer stacked LSTM	0.871	–	–	×
Tripathi et al. (2021a)	Random forest with Feature selection	0.788	0.735	0.764	×
Our Approach	GRU-SMOTE-ENN	0.899	0.931	0.852	✓

patterns that may not be apparent to other ML models. This ability is crucial for predicting credit risk, where past consumer behavior and intricate pattern recognition play a significant role in determining future creditworthiness. Meanwhile, the detailed visualization and analysis of feature impacts, facilitated by SHAP values, enhances transparency, enabling better understanding and trust in the model's predictive decisions. The study emphasizes that the combination of deep learning models with appropriate SMOTE variations like SMOTE-ENN can be a practical and effective approach for managing class imbalance, ensuring reliable and fair credit scoring in real-world financial applications.

6. Conclusion

This study has demonstrated that integrating SMOTE-ENN resampling with deep learning models, particularly GRU, can significantly enhance the accuracy, sensitivity, and specificity of credit risk predictions on imbalanced datasets. Using the Australian and German credit datasets, the GRU-SMOTE-ENN model achieved the highest performance. These findings support the potential of deep learning approaches in financial applications far beyond traditional statistical methods and basic ML models. They also highlight the importance of appropriate data preprocessing to address challenges like class imbalance, which is prevalent in real-world financial data.

The study's outcomes indicate that advanced resampling techniques, when combined with robust deep learning models, can provide a scalable and reliable solution for credit risk assessment, especially in diverse and evolving financial environments. Moreover, the interpretability provided by SHAP values adds another layer of transparency, making these models more acceptable to regulatory bodies and stakeholders, who require clear justifications for automated decisions.

Also, future research should explore the integration of additional data sources, such as transactional behavior and social data, and the use of more complex DL architectures that might further enhance model performance and interpretability. Moreover, studies should continue to refine resampling techniques and explore their effects in various real-world scenarios to ensure that the advances in machine learning are both practical and beneficial in operational settings. Continued emphasis on explainability will also be essential to bridge the gap between sophisticated models and practical deployment in the financial sector.

CRediT authorship contribution statement

Idowu Aruleba: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Visualization. **Yanxia Sun:** Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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